**INTEGRATED SITUATION (UPDATED)**

**Scenario/Problem Description**

KILIKU Cement Limited, located in Bugarama, Rusizi District, Western Province of Rwanda, operates a cement manufacturing plant near the border of the Democratic Republic of Congo.

The company is currently facing serious risks related to **fire outbreaks** in storage areas where cement products are kept. These risks may be caused by:

- Poor monitoring of temperature and smoke
- Lack of early warning systems
- Weak emergency response mechanisms

To address this issue, the company has been advised by an IoT engineer to hire an IoT developer to design and implement a **Fire Detection Security System**.

You are required to design, assemble, and program an **IoT-based Fire Detection System** that detects fire hazards and alerts security personnel.

The task must be completed within **6 hours**.

**GENERAL TASK**

Develop a **web-based system** that:

- Allows **admin registration and login**
- Record and Displays **sensor data (fire detection records)**
- Includes a **logout functionality**

**Task1: Create the following database scheme (10 Marks)**

Database: fire_iot_system

Table 1: admins (**id** (Primary Key, Auto Increment), **name**, **email** (Unique), **password**, **created_at**

Table 2: sensor_data (**id** (Primary Key, Auto Increment), sensor_type,temperature, smoke_level,

flame_detected (1 or 0), status (SAFE / FIRE ALERT), recorded_at)

**Task 2: Registration Page (10 Marks)**

Create a page that:

- Accepts name, email, password

- Stores data in database
- Hashes password
- Prevents duplicate email

**Task 3: Login System (10 Marks)**

Create:

- login.php
- logout.php

Requirements:

- Validate email and password
- Use **sessions**
- Redirect to dashboard on success
- Display error message if login fails

**Task 4: Create Protected Dashboard (10 Marks)**

- Accessible only after login
- Display logged-in admin email
- Include logout link

**Task 5: Display Sensor Data (10 Marks)**

Retrieve data from sensor_data table and display in a table showing:

- Sensor Type
- Temperature
- Smoke Level
- Flame Detected
- Status
- Recorded Time

**Task 6: Insert Sample Data (5 Marks)**

- Insert at least **5 records manually** into the database for testing.

**Task 7: Delete Record (5 Marks)**

- Add a delete button for each record
- Confirm before deleting
- Refresh dashboard after deletion

END